

SAWSAN ABDULBARI

Data Scientist & Machine Learning Engineer | AI Innovation

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BIOGRAPHY

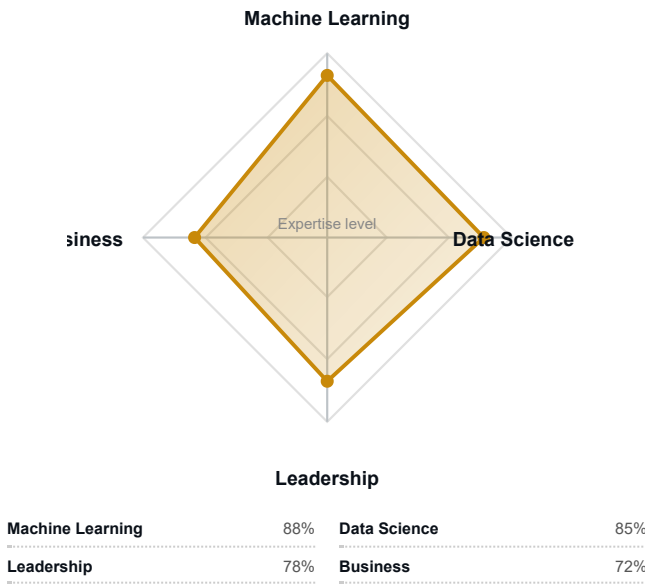
I build full-stack AI and ML systems that ship production machine learning, generative AI, and data software used by real teams to drive measurable business outcomes and automation. Before moving into engineering, I spent over a decade as an operator including 7 years as a founder, plus senior management and quality control roles. That experience shapes how I build AI today: focused on real-world impact, adoption, and audit-ready governance rather than accuracy in isolation. **Trilingual professional** (Arabic, English, Finnish) with **entrepreneurial background** and **passion for climate change solutions**.

Professional overview: I have contributed to environmental sustainability by transforming plastic waste into valuable resources. Throughout my career, I managed supply operations and spearheaded new business initiatives, and focused on developing eco-friendly solutions that align with climate change mitigation efforts.

Commitment to climate change: I am passionate about incorporating climate change awareness into every aspect of business operations and life. My ongoing commitment involves adopting and advocating for practices that minimize environmental footprints, enhance recycling processes, and drive progress toward a more sustainable planet. My approach ensures that business growth and environmental health go hand in hand.

Key Differentiators: Global collaboration (distributed teams in global contributor projects), Production Expertise (deployed ML models), Business Knowledge (7-year entrepreneurial experience), Climate Focus (environmental sustainability advocate), Academic Excellence (student mentor & curriculum developer).

EXPERTISE AND TOPICS



Relative scale aligned with portfolio skills visualization (self-assessment).

Topics: Artificial intelligence, machine learning, data science, generative AI, RAG systems, data analytics, business intelligence, climate change AI.

WORK EXPERIENCE

Data Scientist and Machine Learning Engineer

University of Helsinki | Helsinki, Finland

Anticipated Start: Dec, 2026

Anticipated role: data science and machine learning engineering at the University of Helsinki.

Data Science Machine Learning Python

Junior Expert specializing in AI

Demos Helsinki | Helsinki Metropolitan Area · Hybrid

Aug 2026 - Present

Applying AI expertise to Demos Helsinki: Connected Knowledge System (CKS), contributing to the development and delivery of AI-powered knowledge solutions.

Artificial Intelligence Machine Learning Connected Knowledge System

Data Scientist and Machine learning Engineer

Dashtera | Finland

Jun 2026 - Aug 2026

Built and deployed data science and machine learning solutions, applying end-to-end ML workflows to support Dashtera's products and services.

Data Science Machine Learning Python

Machine Learning Engineer

Omdena | Palo Alto, CA (Remote)

2024 - Present

Collaborating on AI innovation in globally distributed teams. Researching and implementing RAG architecture for enterprise applications.

Python RAG Architecture Machine Learning AI Research

Student Assistant

Häme University of Applied Sciences (HAMK) | Finland

2024 - 2025

Improved educational outcomes for 50+ students and developed curriculum. Provided personalized guidance and technical mentoring.

Python Teaching Curriculum Development

Lead Machine Learning Engineer

Omdena Kitwe, Zambia | Remote

2024

Delivered a COVID-19 forecasting system for the Kitwe, Zambia (Omdena) chapter: XGBoost models, Streamlit app, and standard regression metrics (MAE, RMSE, R²) as in the public repository. Worked with a global team of 15+ contributors.

Team Leadership XGBoost Healthcare AI

Machine Learning Engineer

Raqameyyah Digital Consultancy | Egypt (Remote)

2023 - 2024

Built and deployed ML models using advanced techniques. Developed chatbots with advanced prompt engineering and created front-end applications to enhance customer experience.

Machine Learning Feature Engineering Application Deployment Python

Data Analyst

Raqameyyah Digital Consultancy | Egypt (Remote)

2023

Cleaned and preprocessed large datasets for analysis. Developed applications for data filtering and visualization and created automated data processing pipelines.

Data Analysis Dashboards Streamlit EDA

Founder

Plastic Recycling for Sustainable Solutions | Hybrid

2007 - 2014

Founded and managed a plastic recycling factory focused on environmental protection. Developed business strategies and operational methods and managed employees and production processes. Promoted environmental protection by transforming plastic waste into valuable resources.

Business-to-Business (B2B) New Business Development Supply Operations Environmental Sustainability

EDUCATION

Bachelor of Business Information Technology (Data Science and Machine learning) — Grade 4.7

Häme University of Applied Sciences (HAMK), Finland

2023 - 2026

Specialization in Data Science and Business Intelligence. Focus on machine learning, data analytics, and business applications.

Thesis (BITCA): Microsoft-based performance goal management system for strategy implementation. PDF on the portfolio site. Catalogue: [HAMK Finna \(vanaicat.144542\)](#).

Climate Change AI (CCAI) Summer School

Climate Change AI (CCAI)

2024

Intensive program on machine learning and AI for climate change mitigation, adaptation, and environmental decision-making. Certificate: [Climate_Change_AI_Sawsan_Abdulbari.pdf](#) on the portfolio site.

Google Data Analytics Specialization

Google

Nov 2023

Courses: Foundations: Data, Data, Everywhere; Ask Questions to Make Data-Driven Decisions; Prepare Data for Exploration; Process Data from Dirty to Clean; Analyze Data to Answer Questions; Share Data Through the Art of Visualization; Google Data Analytics Capstone: Complete a Case Study. Certificate PDF: [Google_Data_Analytics.pdf](#) on the portfolio site.

Vocational Qualification in Business (Business Administration)

Tampere Vocational College (Tredu), Finland

2020 - 2022

Business administration and management fundamentals. Project management and organizational skills.

LANGUAGES

Arabic

Native

English

Professional proficiency

Finnish

Professional proficiency

PROJECT REFERENCES

Portfolio — full project collection with live demos [↗](#)

SKILLS

Machine Learning

- Predictive Modeling
- Time Series Analysis
- RAG Architecture
- Model Deployment
- Prompt Engineering

Data Science

- Data Mining & Analytics
- EDA & Visualization
- Feature Engineering
- Statistical Analysis
- Dashboard Creation

Leadership

- Team collaboration (15+ people)
- Cross-cultural Collaboration
- Mentoring & Training
- Project Management
- Stakeholder Communication

Business

- Strategic Planning
- Data Analysis
- Process Optimization
- Financial Management
- Business Development
- Operations Management

SELECTED PUBLICATIONS

Microsoft-based Performance Goal Management System for Strategy Implementation — Bachelor's thesis, Häme University of Applied Sciences (2026). Catalogue: [HAMK Finna \(vanaicat.144542\)](#). PDF on the portfolio site.

PERSONAL INTERESTS

-  **Hiking** - Nature exploration, outdoor activities
-  **Fitness** - Health and wellness, physical activity
-  **Food Exploration** - Culinary experiences, cultural cuisine
-  **Continuous Learning** - Technology trends, professional development
-  **Environmental Protection** - Green initiatives, environmentally friendly practices

Portfolio CV, compiled from the live site at sawsanabdulbari.github.io/portfolio. Links and claims are self-declared.